

OVIS DANIEL IREFU

Power Systems Engineer — Renewable Energy & Grid Reliability

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Authorized to work in the U.S. — no sponsorship required | Open to relocation

PROFESSIONAL SUMMARY

Power Systems Engineer with 10+ years of progressive experience spanning utility distribution operations, renewable energy integration, microgrid design, and revenue analytics. Recovered over \$1.5M in energy losses, improved grid reliability across 17+ substations (comparable to SAIDI/SAIFI reliability benchmarks), and designed/commissioned a hybrid PV-Wind-CSP microgrid testbed. PhD candidate (Dec 2026) combining hands-on utility field leadership with peer-reviewed research in reliability analysis and renewable forecasting. Proficient in MATLAB/Simulink, Python, and Microsoft Excel.

TECHNICAL & RESEARCH COMPETENCIES

Power Systems & Grid: Power Distribution, Load Flow Analysis, Fault Analysis, Substation Operations (17+ substations), Grid Reliability (SAIDI/SAIFI-comparable metrics), ATC&C Loss Reduction

Renewable Energy: Solar PV Design & Optimization, Wind Energy Systems, Battery-Less Off-Grid Systems, Energy Storage (BESS), CSP-Thermoelectric Systems, Renewable Integration & Interconnection

Analysis & Reliability: Reliability Assessment (LPSP, LOLE, ASAI/Availability), Energy Forecasting, Predictive Analytics, Feasibility Studies, Revenue Protection Analysis

Software & Tools: MATLAB/Simulink, Python, Multisim, Arduino, and Advanced Excel, ETAP is currently in active development.

Leadership & Soft Skills: Technical Writing, Field Operations Supervision, Data-Driven Decision Making

EDUCATION

Ph.D. in Electrical Engineering

Aug 2022 – Dec 2026

University of North Texas, Denton, TX | GPA: 3.86/4.00

Dissertation: Design, Implementation & Analysis of a Multi-Source Renewable Microgrid with Hybrid Energy Storage for Enhanced Resilience

M.Eng. in Power Systems and Machines

Apr 2016 – Nov 2019

University of Benin, Edo State, Nigeria | GPA: 3.90/5.00

Thesis: Hybrid PV-Wind-Grid Uninterrupted 2 kW DC Power Supply for Telecom Base Transceiver Stations — 95.9% uptime & renewable energy penetration

B.Eng. in Electrical & Computer Engineering

Nov 2008 – Apr 2014

Federal University of Technology, Minna, Nigeria | GPA: 4.29/5.00

Project: Automatic Wattage Meter using GSM Network for Remote Electricity Billing — reduced non-technical losses by 15%

PROFESSIONAL EXPERIENCE

Graduate Research & Teaching Assistant

Aug 2022 – Present

University of North Texas — Dept. of Electrical Engineering | Denton, TX [UNT](#)

- Designed and implemented a Node-RED-based supervisory monitoring and control architecture for a tri-source off-grid hybrid microgrid integrating solar PV, wind turbine, and concentrating solar power (CSP-TEG) subsystems on a 24V DC bus, enabling real-time visualization and remote management of distributed energy resources.
- Engineered a hardware-software integrated testbed combining embedded sensors, power electronics, and an IoT-enabled dashboard to demonstrate autonomous load management and energy flow optimization in an islanded microgrid configuration.

- Designed, installed, and commissioned a 4.5 kW dual solar PV array + 1 kW wind turbine microgrid at UNT Discovery Park, reducing diesel fuel costs by 60% and accelerating renewable energy adoption for student research and lab operations.
- Engineered a 5-kW battery-less PV-wind off-grid microgrid achieving 95% energy reliability; proposed CSP-hydrogen hybrid configurations, subsequently adopted in UNT's active research roadmap.
- Performed comprehensive reliability analysis (LPSP, LOLE) for a PV-WT microgrid system assessing unplanned outage scenarios and renewable power sharing optimization.
- Developed and delivered EENG 2920 (Analog & Digital Circuit Design) lab curricula, improving student project success rates by 15% and boosting lab engagement scores by 25%.
- Co-authored 4+ peer-reviewed publications on renewable energy systems and microgrid reliability; contributed to 14+ total publications indexed in Google Scholar.

Technical Specialist Supervisor

Mar 2022 – Aug 2022

Abuja Electricity Distribution Plc (AEDC) — Nigeria's federal power distribution utility | Abuja, Nigeria **AEDC**

- Recovered 1.2 MWh in energy theft losses within one month, surpassing 92% of quarterly revenue targets through strategic field interventions, loss audits, and metering upgrades.
- Diagnosed high-loss feeders and streamlined preventive maintenance protocols for 17 substations, reducing unplanned downtime by 20% and improving grid reliability to 93.8% (comparable to SAIDI/SAIFI reliability benchmarks).
- Supervised operation and maintenance of electrical distribution infrastructure including 33kV/11kV switchgear, automatic transfer switches (ATS), and feeder protection relays across 17 substations serving critical commercial and industrial loads.
- Reduced ATC&C losses by 8% in high-risk zones, generating \$102K in cost savings through targeted loss mitigation and network reconfiguration strategies.

Supervisor, Revenue Protection Analyst

Sep 2019 – Mar 2022

Abuja Electricity Distribution Plc (AEDC) | Abuja, Nigeria

- Reduced ATC&C losses by 12% (from 30% to 18%) across 5 districts through data-driven audits, network inspections, and customer energy theft identification — recovering \$926K in FY2020 alone.
- Designed and deployed real-time monitoring dashboards for energy theft detection, improving recovery rates by 16% per month across 4 operational regions.
- Automated meter audit workflows for 10,000+ customers, enhancing billing accuracy by 10% and optimizing field officer deployment.
- Proposed and implemented business process reforms: designed performance dashboards and supervised field teams executing key revenue protection operations.

Energy Data Analyst

Nov 2016 – Sep 2019

Benin Electricity Distribution Plc (BEDC) — Nigerian regional power distribution utility | Benin City, Nigeria **BEDC**

- Reduced ATC&C losses by 10.25% (from 28% to 17.75%) in Etete Business District, saving \$465K over 6 months through predictive analytics, metering audits, and consumption pattern analysis.
- Developed Power BI billing dashboards that increased collection efficiency by 15% and provided management with real-time revenue stream visibility.
- Automated Excel-based reporting systems, cutting data processing time by 30% and improving operational decision-making speed.

Graduate Trainee

Sep 2015 – Oct 2016

Benin Electricity Distribution Plc (BEDC) | Benin City, Nigeria

- Contributed to LV distribution network layout designs and upgraded existing power distribution infrastructure, improving service reliability by 25% for 25,000+ customers in underserved areas.
- Conducted physical inspection tours of LV/HV distribution assets including transformers, switchboards, and metering infrastructure; generated preventive maintenance schedules improving service reliability by 25% for 25,000+ customers.
- Led customer enumeration and asset tagging for 5,000+ meters, achieving 98% inventory accuracy and increasing metering visibility by 35%.

KEY ENGINEERING PROJECTS

- UNT Discovery Park Microgrid (2.5 kW PV + 1 kW Wind + 1.5kW CSP) | PhD Research, UNT | 2022–2026 — Modeled and analyzed battery-less off-grid architecture achieving 95% energy reliability; identified CSP-hydrogen hybrid as optimal enhancement pathway.
- Hybrid PV-Wind-Grid 2 kW BTS Power Supply | M.Eng. Thesis Research | Nigeria | 2016–2019

PUBLICATIONS

- Irefu, O., Ebiega, G., Smithers, B., Acevedo, M. "Reliability Analysis of a Dual Photovoltaic (PV) Microgrid: A Case Study on Renewable Power Sharing and Unplanned Outages"
- Ebiega, G.I., Dugeri, T.M., Irefu, O.D. "Hybrid renewable/grid power systems, an essential for base transceiver station penetration in Rural Nigeria" — DOI: 10.11591/ijpeds.v15.i2.pp1178-1189
- 12 additional peer-reviewed papers in power systems and renewable energy — full list on [Google Scholar](#)

CERTIFICATIONS & PROFESSIONAL AFFILIATIONS

- Graduate Member, Institute of Electrical and Electronics Engineers (IEEE) — No. 99442518
- National Society of Black Engineers (NSBE) — No. 473505
- Corporate Member, Nigeria Society of Engineers (NSE) — No. 47883
- Registered Engineer, Council for the Regulation of Engineering in Nigeria (COREN) — No. R49842
- Junior Management Course, Vigeo Power Training Institute — Oct 2016
- Instrumentation and Control, Petroleum Training Institute — Feb 2015
- Health, Safety and Environment (HSE II and III), Institute of Safety Professionals of Nigeria — Jun 2014